

SITE-TO-SITE VPN (METHOD OF PROCEDURE)

1. Purpose

The purpose of this MOP is to implement a Site-to-Site IPsec VPN between the HQ Network and the Data Center Network across the Internet.

This implementation enables:

- Secure communication between HQ and Data Center resources
- Encrypted transmission of data across an untrusted network
- Bidirectional connectivity between protected networks
- Centralized management of Data Center resources from HQ
- Future expansion of enterprise services into the Data Center
- Secure connectivity while preserving local Internet access at both sites

This change will leverage existing Internet connectivity provided by the HQ and Data Center ISP demarcation routers while introducing a route-based IPsec VPN between HQ-FW-A and DC-FW-A.

2. Scope

This procedure applies to the implementation of a Site-to-Site IPsec VPN between the existing HQ and Data Center environments.

HQ Infrastructure

- HQ-FW-A
- HQ-SW-A
- HQ-HOST-A

Data Center Infrastructure

- DC-FW-A
- DC-SW-A
- DC-HOST-A

Internet Infrastructure

HQ ISP

- WAN Network: 172.16.0.0/29
- ISP Gateway: 172.16.0.1
- HQ-FW-A WAN Interface: 172.16.0.2

Data Center ISP

- WAN Network: 172.30.0.0/30
- ISP Gateway: 172.30.0.1
- DC-FW-A WAN Interface: 172.30.0.2

Protected Networks

HQ Network

- 10.42.0.0/16

Data Center Network

- 10.50.0.0/16

VPN Components

- Phase 1 Configuration
- Phase 2 Configuration
- VPN Policies
- VPN Routes
- NAT Traversal
- Tunnel Validation

3. Pre-Implementation Requirements

Before beginning:

- Confirm administrative access to HQ-FW-A.
- Confirm administrative access to DC-FW-A.
- Verify Internet connectivity from HQ-HOST-A.
- Verify Internet connectivity from DC-HOST-A.
- Verify HQ-FW-A can reach Internet resources.
- Verify DC-FW-A can reach Internet resources.
- Backup both FortiGate configurations.
- Document existing firewall policies and static routes.
- Document HQ and Data Center protected network ranges.

4. Implementation Procedure

Step 0 – Backup Configurations

- Navigate to (do this for both HQ-FW-A and DC-FW-A):
 - Dashboard -> “Username” top right of GUI -> Configuration
 - Select “Backup”
 - Backup to: Local PC
 - Encryption: Yes
 - Password/Confirm Password: “password”
 - Click OK
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Step 1 – Create HQ VPN Tunnel

- Navigate to (in HQ-FW-A):
 - VPN -> IPsec Wizard
 - VPN Setup:
 - Tunnel Name:
 - DC-IPSEC
 - Template:
 - Site to Site
 - Remote Device:
 - FortiGate
 - NAT Between Sites:
 - No NAT
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Step 2 – Configure HQ VPN Authentication

- Authentication:
 - Remote Device:
 - IP Address
 - Remote IP address:
 - 10.3.196.68
 - Outgoing Interface:
 - To_DIA(port8)
 - Authentication method:
 - Pre-shared Key
 - Pre-shared Key:
 - “password”
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Step 3 – Configure HQ VPN Policy & Routing

- Authentication:
 - Local Interface:

- FWA_to_MLS-CORE-A (port2)
 - Local Subnets:
 - Create New:
 - Name: DC-IPSEC_local
 - Type: Subnet
 - Subnet: 10.42.0.0/16
 - Click OK
 - Remote Subnets:
 - Create New:
 - Name: DC-IPSEC_remote
 - Type: Subnet
 - Subnet: 10.50.0.0/16
 - Click OK
 - Internet Access: None
- Review Settings:
 - Click “Create”

Step 4 – Create DC VPN Tunnel

- Navigate to (in DC-FW-A):
 - VPN -> IPsec Wizard
- VPN Setup:
 - Tunnel Name:
 - HQ-IPSEC
 - Template:
 - Site to Site
 - Remote Device:
 - FortiGate
 - NAT Between Sites:
 - No NAT

Step 5 – Configure DC VPN Authentication

- Authentication:
 - Remote Device:
 - IP Address
 - Remote IP address:
 - 10.3.196.26
 - Outgoing Interface:
 - ISP-DC(port8)
 - Authentication method:
 - Pre-shared Key
 - Pre-shared Key:
 - “password”

Step 6 – Configure DC VPN Policy & Routing

- Authentication:
 - Local Interface:
 - DC_SERVERS
 - Local Subnets:
 - Create New:
 - Name: HQ-IPSEC_local
 - Type: Subnet
 - Subnet: 10.50.0.0/16
 - Click OK
 - Remote Subnets:
 - Create New:
 - Name: HQ-IPSEC_remote
 - Type: Subnet
 - Subnet: 10.42.0.0/16
 - Click OK
 - Internet Access: None
 - Review Settings:
 - Click “Create”
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Step 7 – Create HQ VPN Policies

- Navigate to (in HQ-FW-A):
 - Policy & Objects -> Firewall Policy
 - Create New:
 - Name:
 - HQ_TO_DC
 - Incoming Interface:
 - FWA_to_MLS-CORE-A(port2)
 - Outgoing Interface:
 - DC_IPSEC
 - Source:
 - 10.42.0.0/16
 - Destination:
 - 10.50.0.0/16
 - Schedule:
 - Always
 - Action:
 - Accept
 - NAT:
 - Disabled
 - Select OK
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Step 8 – Create DC VPN Policies

- Navigate to (in DC-FW-A):
 - Policy & Objects -> Firewall Policy
 - Create New:
 - Name:
 - DC_TO_HQ
 - Incoming Interface:
 - DC_SERVERS
 - Outgoing Interface:
 - HQ_IPSEC
 - Source:
 - 10.50.0.0/16
 - Destination:
 - 10.42.0.0/16
 - Schedule:
 - Always
 - Action:
 - Accept
 - NAT:
 - Disabled
 - Select OK

Step 9 – Create HQ Static Routes

- Navigate to (in HQ-FW-A):
 - Network -> Static Routes
 - Create New:
 - Destination
 - 10.50.0.0/16
 - Interface:
 - DC-IPSEC
 - Select OK

Step 10 – Create DC Static Routes

- Navigate to (in DC-FW-A):
 - Network -> Static Routes
 - Create New:
 - Destination
 - 10.42.0.0/16
 - Interface:
 - HQ-IPSEC
 - Select OK

5. Validation Plan

Step 1 – Verify VPN Status

- Navigate to (in HQ-FW-A):
 - VPN -> IPsec Tunnels
 - Confirm:
 - DC-IPSEC
 - Status: Up
 - Navigate to (in DC-FW-A):
 - VPN -> IPsec Tunnels
 - Confirm:
 - HQ-IPSEC
 - Status: Up
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Step 2 – Verify Phase 2 Status

- In HQ-FW-A CLI:
 - Run: `> get vpn ipsec tunnel summary`
 - Confirm:
 - Selectors (total,up): 1/1
 - In DC-FW-A CLI:
 - Run: `> get vpn ipsec tunnel summary`
 - Confirm:
 - Selectors (total,up): 1/1
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Step 3 – Verify HQ > DC Connectivity

- From HQ-HOST-A
 - Run: `> ping 10.50.10.10`
 - Expected Result: Successful ICMP replies
 - From DC-HOST-A
 - Run: `> ping 10.42.10.6`
 - Expected Result: Successful ICMP replies
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Step 4 – Verify VPN Traffic Counters

- In HQ-FW-A CLI:
 - Run: `> get vpn ipsec tunnel summary`
 - Expected Result: RX/TX Packet Count Increases

- In DC-FW-A CLI:
 - Run: `> get vpn ipsec tunnel summary`
 - Expected Result: RX/TX Packet Count Increases
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6. Rollback Procedure

Rollback should be performed if:

- Phase 1 fails to establish
 - Phase 2 fails to establish
 - Protected networks cannot communicate
 - Validation testing fails
 - Unexpected routing behavior is observed
 - Network instability occurs following implementation
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Step 1 – Restore Configurations

- Navigate to (do this for both HQ-FW-A and DC-FW-A):
 - Dashboard -> “Username” top right of GUI -> Configuration
 - Select “restore”
 - Upload backup file
 - Password: “password”
 - Click OK
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Step 2 – Validate Rollback

- Confirm:
 - DC-IPSEC no longer exists
 - HQ-IPSEC no longer exists
 - Routes/networks no longer exist